SAMUEL PETER

Software, DevOps & Cloud Engineer | Critical Infrastructure, AI & Reliability

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PROFESSIONAL SUMMARY

Software, DevOps and Cloud Engineering practitioner pursuing an MSc in DevOps & Cloud Computing, with First Class Honours in Computing and **20+ years leading national electricity-grid operations**. Combines critical-infrastructure reliability, incident response and operational governance with hands-on React, TypeScript, Python, cloud, CI/CD and API engineering. Developed **GridPulse Enterprise** and **PowerPulse**, two globally oriented AI-assisted energy platforms spanning TSO decision support and human-centered household utility intelligence. Focused on dependable, secure and maintainable software for enterprise engineering teams.

TECHNICAL SKILLS

Programming & AI: TypeScript, JavaScript, Python, Java, React, Node.js, Express, FastAPI, REST APIs, OpenAPI, Google Gemini AI
Cloud & delivery: AWS, Docker, Kubernetes, Terraform, Git, GitHub Actions, Jenkins, Maven, Vercel, Supabase, PostgreSQL
Quality & operations: JUnit, Selenium, k6, CI/CD, automated validation, monitoring, incident response, RCA, EMS/SCADA, telemetry simulation

SELECTED SOFTWARE ENGINEERING PROJECTS

GridPulse Enterprise | Global TSO SCADA/EMS & AI Dispatch Intelligence

React 19 | TypeScript | Vite | Express | Google Gemini | Vercel

- Developed a multi-region TSO operations simulator with configurable grid standards, SCADA/EMS telemetry, contingency alerts and sequence-of-events logging.
- Integrated modeled weather intelligence, transformer DGA and asset diagnostics with AI-assisted dispatch recommendations and simulated remedial actions.

Repository: github.com/sampeter-akan/gridpulse-330app

PowerPulse | Global Utility Intelligence & Human-Centered Energy

React | TypeScript | Node.js | Express | Google Gemini | Vercel

- Designed an IDEO human-centered, country-aware utility platform with local currencies, provider contexts, energy forecasts, appliance demand response and solar planning.
- Implemented AI-assisted meter interpretation, tariff auditing, community outage awareness and health-sensitive safeguards; tested with real household meter data in Nigeria.

Repository: github.com/sampeter-akan/powerpulse

VAULT FIVE | Real-Time Multiplayer Web Application

TypeScript | Vite | Supabase Edge Functions | PostgreSQL | Realtime | GitHub Actions

- Built a browser-based room-code multiplayer game with anonymous access, synchronized rounds and serverless orchestration; documented CI checks and operational runbooks.

Repository: github.com/sampeter-akan/vault-five

Movies API | Contract-First Microservices

TypeScript | Node.js | OpenAPI v3 | k6 | GitHub Actions

- Built contract-first REST services with automated validation and CI/CD; used repeatable k6 performance tests to examine endpoint behavior.

Repository: github.com/sampeter-akan/movies-api

Credit Loan Calculator | Automated Build & Test Pipeline

Java | Jenkins | Maven | Selenium | JUnit | GitHub Actions

- Implemented automated build and test stages, dependency update automation and structured logging to support maintainable delivery.

Repository: github.com/sampeter-akan/credit-loan-calculator

HomeStayGrid | ML Recommender Feed

Python | FastAPI | Docker | AWS ECR | Elastic Beanstalk

- Built and deployed a containerized ML recommender prototype for maintenance prioritization using AWS deployment workflows.

Repository: github.com/sampeter-akan/homestaygrid_ml_recommender_feed

PROFESSIONAL EXPERIENCE

Transmission Company of Nigeria (TCN) | Grid System Operations | Jun 2005–Present

Principal Manager (Jul 2025–Present) | **Senior Manager** (Jul 2021–Jun 2025) | **Manager** (Jul 2018–Jun 2021)

- Lead monitoring, incident response, operational risk and change governance in national high-voltage transmission operations; support service availability reported at 99.9%+.
- Directed incident analysis and corrective actions associated with approximately 30% lower outage impact; trained 15+ operators and developed operating runbooks.

Assistant Manager (Jan 2015–Jun 2018) | **Officer I** (Jul 2011–Dec 2014) | **Officer II** (Jun 2005–Jun 2011)

- Progressed through 24/7 control-room operations, switching coordination, event tracking, escalation, restoration support and operational documentation.

EDUCATION

MSc DevOps & Cloud Computing | IU International University of Applied Sciences, Germany | In progress

GPA: 80.5% | DevOps Engineering: 96% | Research Methods: 90% | Cloud Computing Project: 84% | Project: Design Thinking: 95%

BSc (Hons) Computing & Information Technologies | University of Derby, UK | 2020 | First Class Honours

BTEC Level 5 HND, Computing and Systems Development | Pearson RDI, UK | 2014

LANGUAGES

English (C1) | Efik/Ibibio (Native)